

INTEGRATIVE PROJECT OF THE 5TH SEMESTER OF LEI-ISEP

2025-2026 (version I.b)

PART I – Context and Operation

1. Context

In the academic year 2025-2026, the fifth semester (i.e., the 3rd year, 1st semester) of the Bachelor's Degree in Informatics Engineering (LEI) at the Instituto Superior de Engenharia do Porto (ISEP) adopts a teaching-learning process based on the development of a single project that enhances the integration and application of knowledge, skills and competencies of all course units (UCs) taught through the semester: Administração de Sistemas (ASIST), Arquitetura de Sistemas (ARQSI), Gestão (GESTA), Inteligência Artificial (IART), Laboratório e Projeto V (LAPR5) e Sistemas Gráficos e Interação (SGRAI). The project, common to all course units, follows the schedule outlined in Table 1.

TABLE 1. SEMESTER AND PROJECT CALENDAR.

<i>Sprint</i>	Begins		Ends		General Goal
	Week	Day	Week	Day	
A	1	15/09/2025	6	26/10/2025	Project set up and teams' formation. Development of the 1 st Iteration (MVP).
B	7	27/10/2025	10	23/11/2025	Development of the 2 nd Iteration (MVP).
C	11	24/10/2025	14	21/12/2025	Development of the 3 rd Iteration (MVP).
-	Christmas Holidays				
-	15	05/01/2026	15	09/01/2026	Final Assessment to all UC.

To develop the project, students must form groups/work teams common to all course units. The goal is to obtain an integrated solution, that is, a single system encompassing the various components / applications developed.

The operating rules outlined in this document (cf. section 2) are guidelines covering the main scenarios. Other scenarios, distinct of those mentioned in this document, must be presented to the responsible (RUC) of LAPR5, who, together with the RUCs of the other course units, will decide how to proceed.

It is important to emphasize that the project development must demonstrate both (i) the best practices, tec acquired throughout previous semesters; and (ii) the best practice, techniques and technologies acquired/developed throughout this semester. The application of these best practices, techniques and technologies is an integral part of the final project assessment criteria. The system to be developed is described in a separate document with the same title as this one and whose subtitle is "PART II – System Specification."

2. Operation Mode

The development of the project includes the participation and application of the knowledge and skills acquired in all the semester's UC in an integrated and phased manner, as presented in Table 2 (the "X" indicates that the UC participates in the respective project sprint).

TABLE 2. PARTICIPATION OF EACH UC IN THE PROJECT DEVELOPMENT.

<i>Sprint</i>	LAPR5	ARQSI	GESTA	ASIST	IART	SGRAI
A	X	X	X			
B	X	X	X	X	X	X
C	X	X	X	X	X	X

The project will be developed primarily during the lab (PL) classes, or practical (TP) classes, of each course unit (including LAPR5), which will be dedicated to the application/problem-solving, best practices and techniques related to the UC's syllabus. Additionally, and according to each team's needs, the project may (and is expected to) be developed outside of classes.

2.1. Team Composition

The project is envisioned to be carried out by teams of **4 students**, who must remain the **same across all UCs** and preferably all students belonging to the same lab (PL) class¹. Each team operates as it belongs to a company that is competing against the others to develop the desired system. **Each team's focus should always be on the project as a whole and not on the scope of each UC individually.**

Ideally, all team members (students) should be enrolled in all UC (see Table 3). However, given the diversity of LEI students, other team compositions may be acceptable, as long as each course unit has at least three students enrolled (cf. Table 4 example).

TABLE 3. IDEAL SCENARIO FOR TEAM COMPOSITION.

	ASIST	ARQSI	GESTA	IART	LAPR5	SGRAI
Student 1	X	X	X	X	X	X
Student 2	X	X	X	X	X	X
Student 3	X	X	X	X	X	X
Student 4	X	X	X	X	X	X
Students/UC	4	4	4	4	4	4

TABLE 4. EXAMPLE OF A TEAM COMPOSITION MEETING THE MINIMUM ENROLLMENT CRITERIA.

	ASIST	ARQSI	GESTA	IART	LAPR5	SGRAI
Student 1		X	X	X	X	X
Student 2	X		X	X	X	X
Student 3	X	X		X		X
Student 4	X	X	X		X	
Students/UC	3	3	3	3	3	3

¹ Regarding GESTA, the students must be of the same practical (TP) class.

Other cases than those mentioned in this document should be presented to the responsible (RUC) of LAPR5, who, together with the RUCs of the other course units, will decide how to proceed. It is suggested that students form teams during the lab (PL) classes of the UCs taking place in the first weeks and, at that time, request pre-validation/acceptance of the team composition from the respective class instructor. If approved, **a team member must submit the team composition through the form available on the fifth semester Integrative Project page on Moodle**. Exceptionally, the team composition may be submitted via email to the RUC of LAPR5.

2.2. Team Working Methodology

At the beginning of each sprint, a sprint backlog is provided. It consists of a set of user stories specifying the work to be performed by the team during the sprint. Then, the team is responsible for (per sprint):

- a) analyzing this information and, if necessary, **requesting clarification in a timely manner for the successful completion of these user stories;**
- b) **correctly applying the engineering process**, particularly the software development process and its various activities (e.g., analysis, design, implementation, testing, deployment), **leaving evidence of its practice;**
- c) **planning the tasks** to be performed considering both the sprint objectives and the existence of possible dependencies between tasks and, whenever necessary, reviewing and modifying the previously prepared plan;
- d) **ensuring a balanced division of tasks and responsibilities among all team members** (e.g., in terms of effort, knowledge, and complexity) such that **any team member is able to describe and explain how a given requirement was/is implemented/met in the system and, if requested, is also capable of adding improvements/changes (e.g., when conducting an assessment).**

Be aware that it is neither intended nor acceptable having any team member dedicate themselves solely to tasks that require the application of knowledge from a single course unit (e.g., ARQSI or SGRAI only). Rather, each team member should participate in tasks that require the application of knowledge from all courses units in which they are enrolled.

It is also important emphasizing that everyone is responsible for ensuring the smooth running of the team. In the event of conflicts, the team should first attempt to resolve the conflict within the team and as a team. If this is not enough to solve the issue, the team should report it to the faculty responsible for monitoring the team.

Finally, it is recommended that the team work regularly (e.g., daily, weekly) on the project and adopt appropriate tools, processes, and methods for effective project management, collaboration and execution, as well as for sharing and communicating information among all team members.

2.3. Project Assessment

At the end of each sprint, each participating course unit will conduct an individual and/or team assessment (formative and/or summative) of the work completed so far. **The assessment criteria are defined exclusively by each course unit and generally focus on the technical aspects of the project that fit within its syllabus. Therefore, the criteria applied vary and differ from one course unit to another.** The results (grades) obtained in one course unit do not directly influence the results of the other course units. Therefore, it is extremely important that students apply an appropriate work

methodology and, in particular, regarding the tasks assignment (see section 2.2) to prevent any undesirable consequences (e.g., failure to achieve the objectives in a given course unit).

At last, it should be notice that every course unit assessment will be performed in accordance with what is established in the respective Course Syllabus (FUC) and may include other assessments than the ones upcoming from the project development. In case of a given UC having additional assessments, those will tend to occur before the UC beginning its participation in the project or after the project development has ended (i.e., during the examination period).